LOCALIZING TWICE ALONG A MORPHISM OF THE SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Composition of cocontinuous functors. Same chapter, Lemma lemma-composition-cocontinuous. Gives the commutativity of the diagram of morphisms of topoi attached to the composite functor.
- The localization of a site in an object. Same chapter, the definition of the localization functor and of the associated morphism of topoi. Supplies the functors whose composites are computed.

Source reference. Original lemma 03L5: sources/stacks-sites/pinned-original-source.tex, source lines 5697–5697. Navigation pointer invoked at proof line 5697; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $f : V \to U$ be a morphism of $\mathcal{C}$. Then there exists a commutative diagram*

$$\begin{array}{ccc} \mathcal{C}/V & \xrightarrow{j} & \mathcal{C}/U \\ & {\scriptstyle j_V} \searrow \quad \swarrow {\scriptstyle j_U} & \\ & \mathcal{C} & \end{array}$$

of continuous and cocontinuous functors. The functor $j : \mathcal{C}/V \to \mathcal{C}/U$, $(a : W \to V) \mapsto (f \circ a : W \to U)$ is identified with the functor $j_{V/U} : (\mathcal{C}/U)/(V/U) \to \mathcal{C}/U$

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via the identification $(\mathcal{C}/U)/(V/U) = \mathcal{C}/V$. *Moreover we have* $j_{V!} = j_{U!} \circ j_!$, $j_V^{-1} = j^{-1} \circ j_U^{-1}$, *and* $j_{V*} = j_{U*} \circ j_*$.

Proof. The commutativity of the diagram is immediate. The agreement of j with $j_{V/U}$ follows from the definitions. By Lemma 03L5 we see that the following diagram of morphisms of topoi

$$\begin{array}{ccc} Sh(\mathcal{C}/V) & \xrightarrow{\quad j \quad} & Sh(\mathcal{C}/U) \\ & {\scriptstyle j_V} \searrow \quad \swarrow {\scriptstyle j_U} & \\ & Sh(\mathcal{C}) & \end{array} \tag{0.1.1}$$

is commutative. This proves that $j_V^{-1} = j^{-1} \circ j_U^{-1}$ and $j_{V*} = j_{U*} \circ j_*$. The equality $j_{V!} = j_{U!} \circ j_!$ follows formally from adjointness properties. □